

Commerce XI Maths App

Exercise 2.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 13

Question 1 - Q1(i)

For 7, 11, 2, 4, 9, 6, 3, 7, 11, 2, 5, 8, 3, 6, 8, 8, 2, 6
Calculate variance.

Correct answer: 8

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 8.

Question 2 - Q1(ii)

For 7, 11, 2, 4, 9, 6, 3, 7, 11, 2, 5, 8, 3, 6, 8, 8, 2, 6
Calculate standard deviation.

Options

A. $2\sqrt{2}$

B. 8

C. 4

D. $\sqrt{2}$

Correct answer: $2\sqrt{2}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $2\sqrt{2}$.

Question 3 - Q2(i)

For 65, 77, 81, 98, 100, 80, 129
Calculate variance.

Correct answer: 380

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 380.

Question 4 - Q2(ii)

For 65, 77, 81, 98, 100, 80, 129
Calculate standard deviation.

Options

A. $2\sqrt{95}$

B. 380

C. $\sqrt{95}$

D. 19

Correct answer: $2\sqrt{95}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $2\sqrt{95}$.

Question 5 - Q3(i)

Compute variance for the following data:

x	2	4	6	8	10
f	5	4	3	2	1

Correct answer: 6.1911

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 6.1911.

Question 6 - Q3(ii)

Compute standard deviation for the following data:

x	2	4	6	8	10
f	5	4	3	2	1

Options

A. $\sqrt{6.1911}$

B. 6.1911

C. $\sqrt{5.1911}$

D. $\sqrt{7.1911}$

Correct answer: $\sqrt{6.1911}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{6.1911}$.

Question 7 - Q4(i)

Compute variance for the following data:

x	1	3	5	7	9
Frequency	5	10	20	10	5

Correct answer: 4.8

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 4.8.

Question 8 - Q4(ii)

Compute standard deviation for the following data:

x	1	3	5	7	9
Frequency	5	10	20	10	5

Options

A. $\sqrt{(4.8)}$

B. 4.8

C. $\sqrt{(3.8)}$

D. $\sqrt{(5.8)}$

Correct answer: $\sqrt{(4.8)}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{(4.8)}$.

Question 9 - Q5(i)

Following data gives age of 100 students in a school. Calculate variance

Age (in years)	10	11	12	13	14
No. of students	10	20	40	20	10

Correct answer: 1.2

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 1.2.

Question 10 - Q5(ii)

Following data gives age of 100 students in a school. Calculate S. D.

Age (in years)	10	11	12	13	14
No. of students	10	20	40	20	10

Options

A. $\sqrt{(1.2)}$

B. 1.2

C. $\sqrt{(0.2)}$

D. $\sqrt{(2.2)}$

Correct answer: $\sqrt{(1.2)}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{(1.2)}$.

Question 11 - Q6

The mean and variance of 5 observations are 3 and 2 respectively.

If three of the five observations are 1, 3 and 5,

find the values of other two observations.

Options

A. 2, 4

B. 1, 5

C. 2, 5

D. 3, 4

Correct answer: 2, 4

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 2, 4.

Question 12 - Q7

Obtain S. D. for the following data :

Height (in inches)	60-62	62-64	64-66	66-68	68-70
No. of students	4	30	45	15	6

Options

A. $\sqrt{(3.3514)}$

B. 3.3514

C. $\sqrt{(2.3514)}$

D. $\sqrt{(4.3514)}$

Correct answer: $\sqrt{(3.3514)}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{(3.3514)}$.

Question 13 - Q8

The following distribution was obtained by change of origin and scale of variable X.

d_i	-4	-3	-2	-1	0	1	2	3	4
f_i	4	8	14	18	20	14	10	6	6

If it is given that mean and variance are 59.5 and 413 respectively, determine actual class intervals.

Options

- A. 15.5–25.5, 25.5–35.5, ..., 95.5–105.5
- B. 19.5–29.5, 29.5–39.5, ...
- C. -4.5–4.5, 4.5–14.5, ...
- D. 55.5–65.5, 65.5–75.5, ...

Correct answer: 15.5–25.5, 25.5–35.5, ..., 95.5–105.5

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 15.5–25.5, 25.5–35.5, ..., 95.5–105.5.